Technical supplement

GMO for multicategory questions

GM and GMO give the same results on the 101 questions with more than two answers. For GMO, I used the vector-odds extension in the article “When pooling forecasts, use the geometric mean of odds”. Both methods calculate a geometric average for each answer, then rescale the averages to total 100%. They also agree when given the same aging weights. The methods differ only on Yes/No questions.

Treatment of very small probabilities

A single zero makes a geometric average zero. Very small probabilities also pull it down strongly. I limited this influence by raising every probability below 0.1% to 0.1% before calculating GM and GMO. I raised small positive values as well as zeros. Otherwise, a 0% forecast would become 0.1%, higher than an unchanged 0.01% forecast.

Raising these values makes the probabilities total more than 100%. I then reduced all probabilities proportionally to restore the total.

I repeated the half-life comparison with minimum probabilities of 0.01% and 1% to see whether this choice affected the results. The table shows how much daily halving reduced error compared with no decay. I also checked which half-life gave the lowest error at each minimum.

Minimum before rescaling	GM error reduction	GMO error reduction	Best tested half-life for both
0.01%	12.1%	13.7%	1 day
0.1%	12.6%	14.2%	1 day
1%	13.6%	15.0%	0.5 day

Daily halving reduced error at every minimum. The best tested half-life changed at 1%, so these results support aging weights without settling on one exact rate.

Uncertainty in score differences

For each question, I compared its average Brier score with and without daily halving. I averaged these differences across all 187 questions and calculated the standard error, which measures uncertainty in this average improvement.

For GM, the average reduction was 0.0338, with a standard error of 0.0071 (21% of the reduction). For GMO, it was 0.0391, with a standard error of 0.0077 (20%).

I also compared GM and GMO with daily halving. GM's average score was 0.0012 lower, with a standard error of 0.0086, about seven times the difference. This difference is not statistically significant.

These standard errors assume that the questions are independent. They do not account for choosing the decay rate using these same results. I would check the gains on new questions.

Selecting the decay rate

I tested a rule that chooses the decay rate from past results. It could choose no decay or a half-life of 1, 3, 7, 14, 30 or 60 days.

Each day, the rule compared these same options on questions whose answers were already known by the previous day's cutoff. More questions entered the comparison as their answers became known. It used the option with the lowest average error. Until the first answer was known, it used no decay.

After that, the rule always chose daily halving for both GM and GMO. GM's overall score was 0.2351, a reduction of 12.4% from no decay. GMO's was 0.2364, a reduction of 14.0%. I chose which rates to test after exploring the data, so we still want to test on new questions.

Next tests

For the following tests, I would use one set of questions to choose the settings and a separate set to evaluate them. I would compare GM, GMO and the arithmetic mean across the same 12 half-lives from 0.01 to 60 days, and with no decay. I would plot the scores to see which option performs best for each method.

In the current analysis, the arithmetic mean had its lowest tested score at half a day: 0.2456. This was still higher than GM and GMO at one day, with scores of 0.2345 and 0.2357. I would also apply aging weights to the median and trimmed mean. All methods would have the same half-lives to choose from and the same procedure for choosing a rate.

As a next exploratory step, I would compare the fixed minimum with a quantile treatment. That treatment replaces exact 0% and 100% using lower and upper percentiles of the other forecasts. The replacement values therefore depend on the forecasts being combined. For each treatment, I would calculate GM directly from the Yes probabilities and then directly from the No probabilities, since this choice and the treatment of small probabilities can interact. I would check how the scores change and whether aging weights still help.

Newer forecasts may contain new information or come from more accurate forecasters. I would compare four rules on the same forecasts: equal, constant weights; aging weights; weights based on past accuracy; and weights that combine age and past accuracy. Comparing the combined rule with each rule alone would help determine whether age and past accuracy each contribute to the improvement.

To measure past accuracy, I would use only outcomes already known at the time. I would compare accounts on the same questions and dates. Otherwise, an account could appear more accurate because it forecast more predictable events or forecast later, when more information was available.